

Logarithmic functions

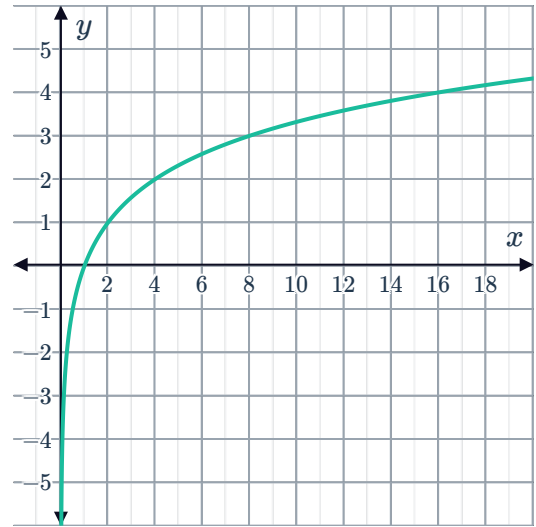


1

a

x	$\frac{1}{2}$	1	2	4	16
y	-1	0	1	2	4

b

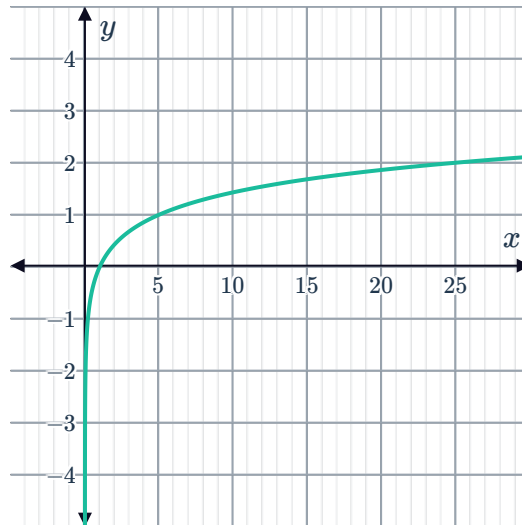


c $x = 0$

d

- i No ii No iii Yes iv No
v No

2



3



a

x	$\frac{1}{243}$	$\frac{1}{3}$	1	3	9	81
y	-5	-1	0	1	2	4

b Increasing

c $y = \log_3 x$ approaches $-\infty$ as x approaches zero.d y is undefined.

Inverses of logarithmic functions

4

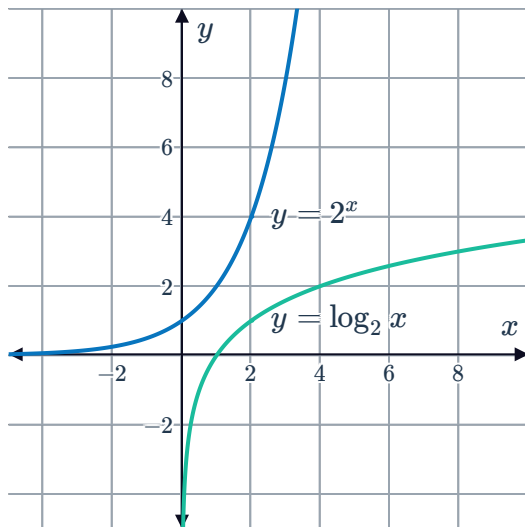
a

x	1	2	4	8
y	0	1	2	3

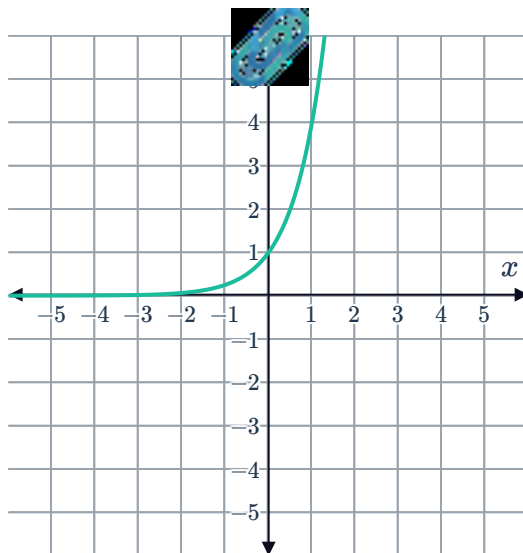
b

x	0	1	2	3
y	1	2	4	8

c

d $y = 2^x$

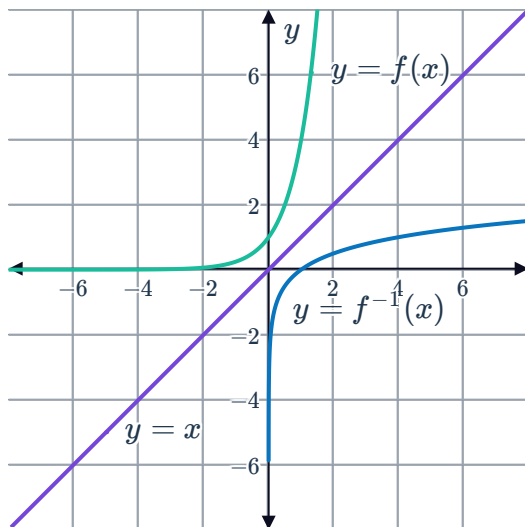
5



6

a

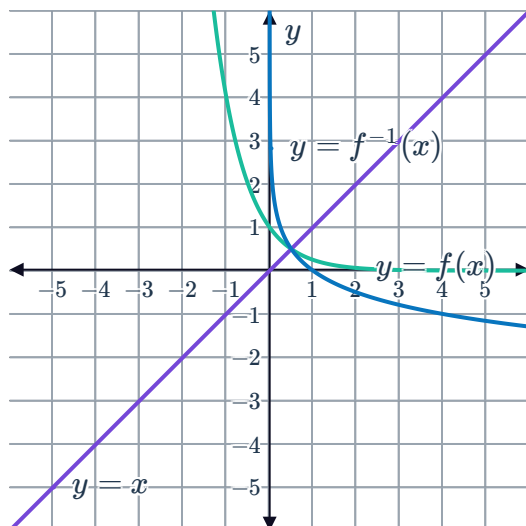
i



ii $y = \log_4 x$

b

i

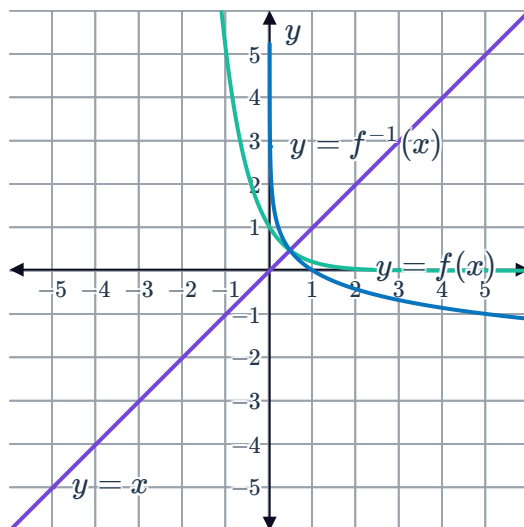


ii $y = -\log_4 x$

c
i



ii $y = \log_{\frac{1}{5}} x$

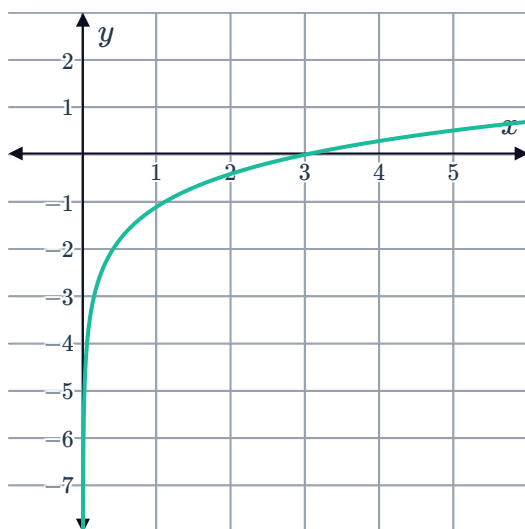


Natural logarithmic functions

7 The logarithmic function with base e .

8

a



b $y = \ln \left(\frac{x}{3} \right)$

9

a $y = \log_e \left(\frac{x+5}{2} \right)$

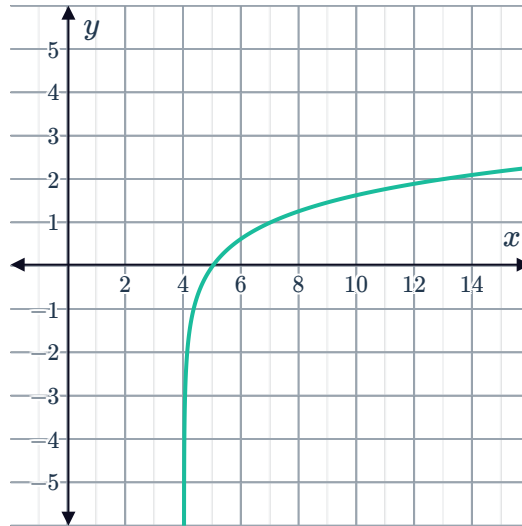
b $(-4.986, -4.986), (1.118, 1.118)$

Transformations of logarithmic functions



10 $y = \log_3 (x - 5)$

11

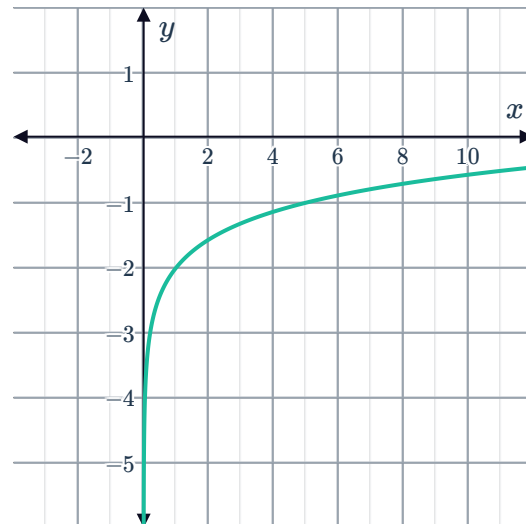


12

a

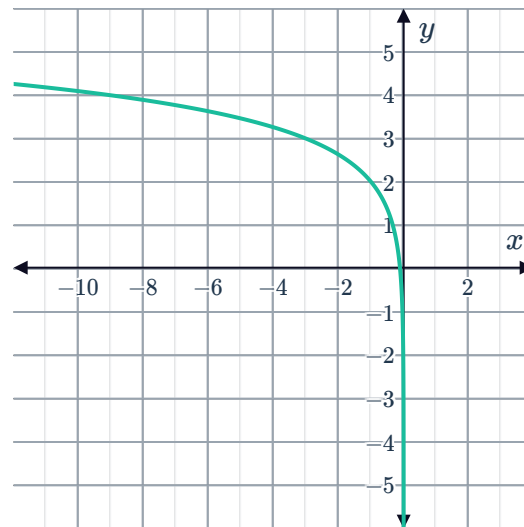
i $y = \log_5 x - 2$

ii



b

i $y = \log_3(-x) + 2$



13 $g(x) = 3 \log_5 x$

14

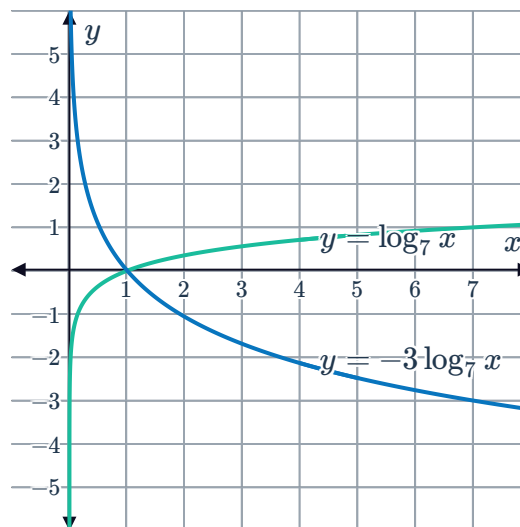
a Reflection over the y -axis.

b $g(x) = \ln(-x)$

15

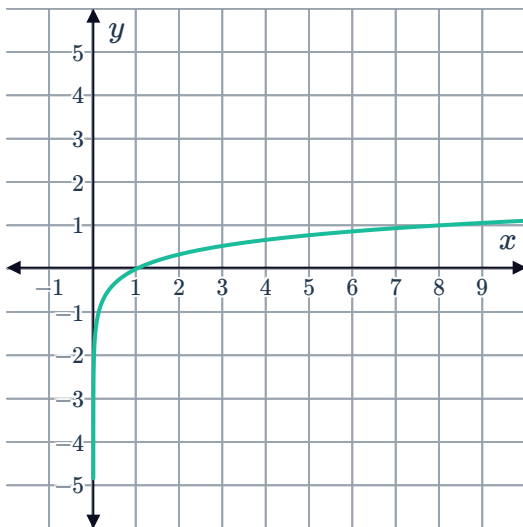
a Vertical stretch by a factor of 3 and reflection across the x -axis.

b

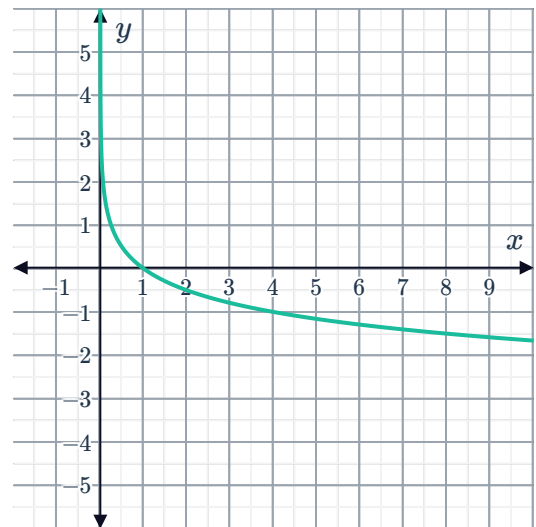


16

a



b



17

a

x	-4	-1	$-\frac{1}{4}$
$\log_4(-x)$	1	0	-1

b

x	-2	1	$\frac{7}{4}$
$\log_4(2-x)$	1	0	-1

c $y = \log_4(2-x)$ is a translation of $y = \log_4(-x)$ by 2 units to the right.

18

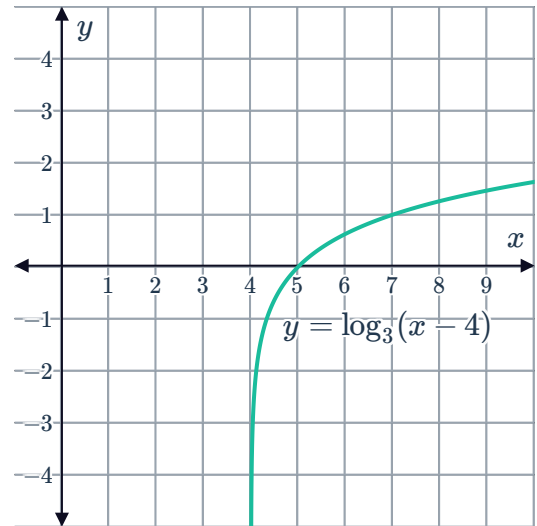
a $x = 4$

c 1



b $(5, 0)$

d



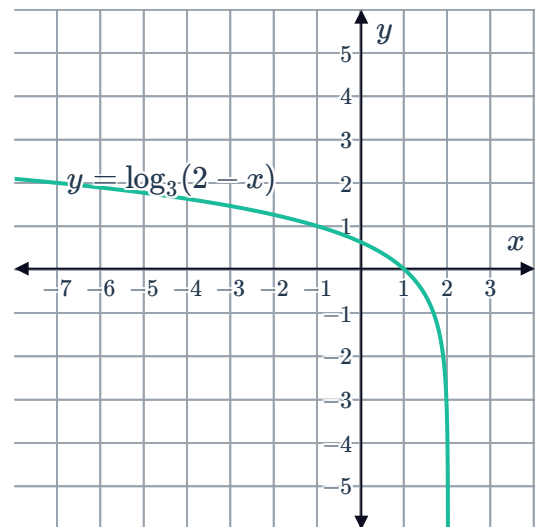
19

a $x = 2$

c 2

b $(1, 0)$

d



20

a $x = 0$

e $x = -3$

b $x = 0$

f $x = 0$

c $x = 5$

d $x = 0$

21

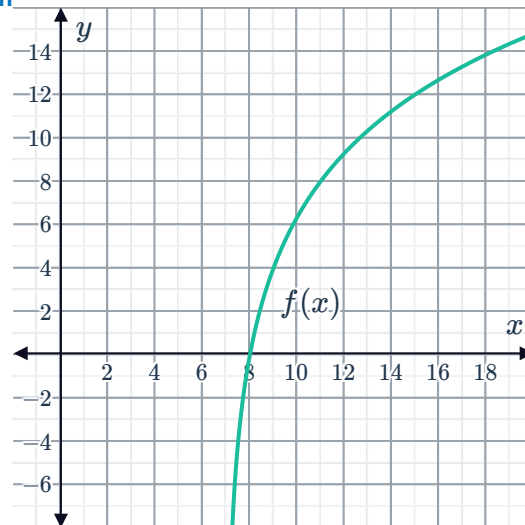
a

i $x = 8$

ii $x = 7$



iii

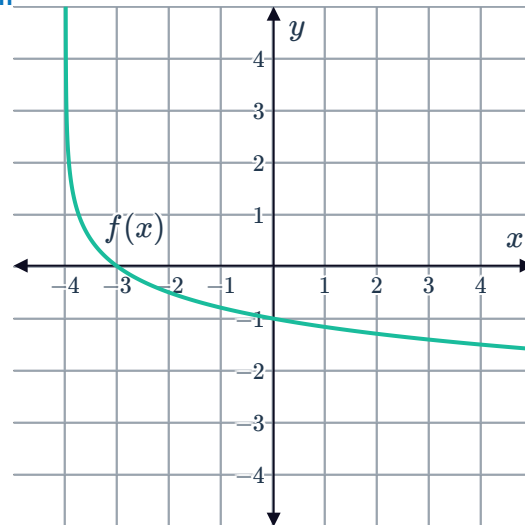


b

i $x = -3$

ii $x = -4$

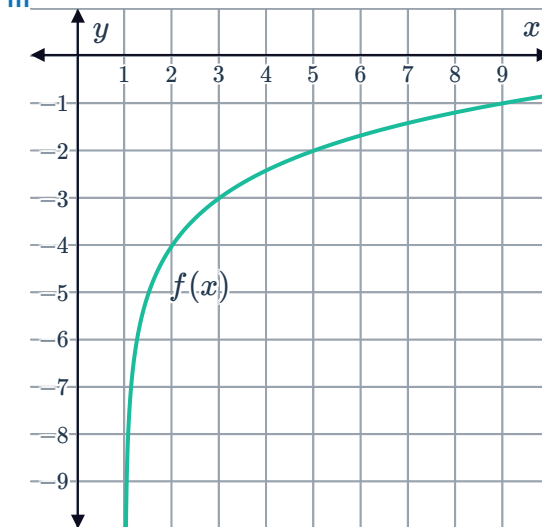
iii



c

i $x = 17$

ii $x = 1$

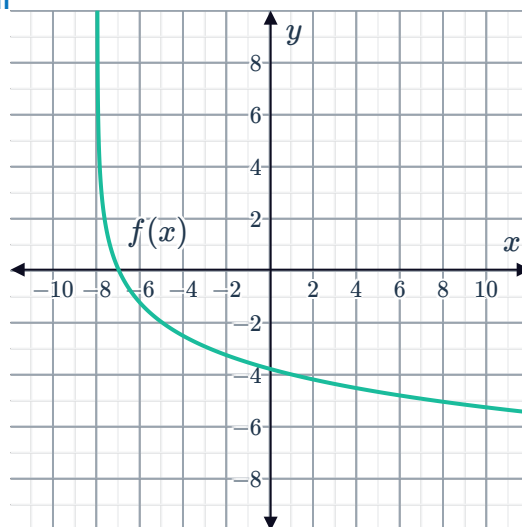


d

i $x = -7$

ii $x = -8$

iii

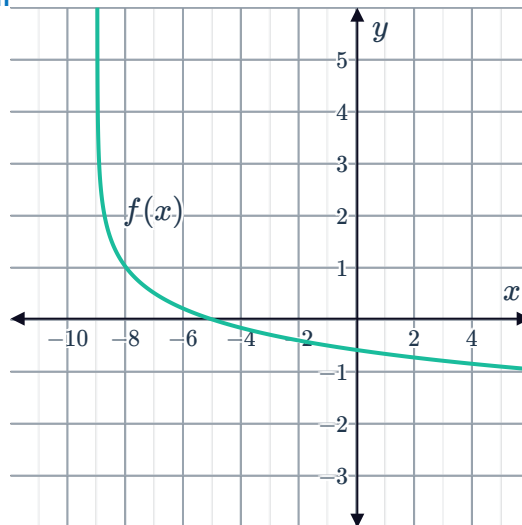


e

i $x = -5$

ii $x = -9$

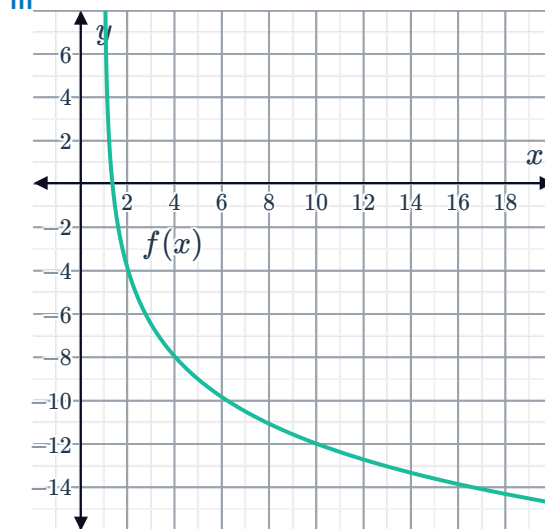
iii



f

i $x = \frac{4}{3}$

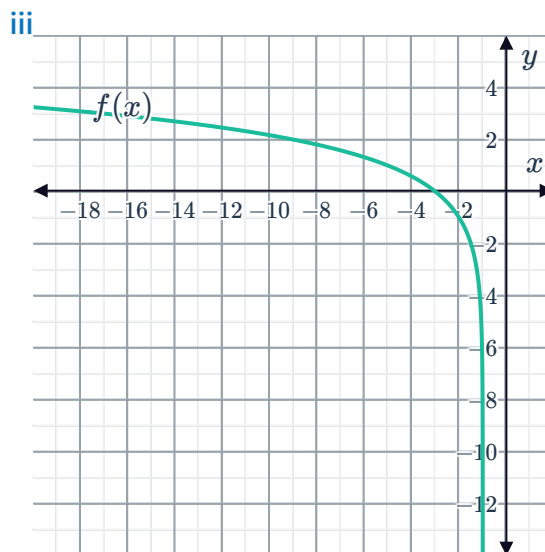
ii $x = 1$



g

i $x = -3$

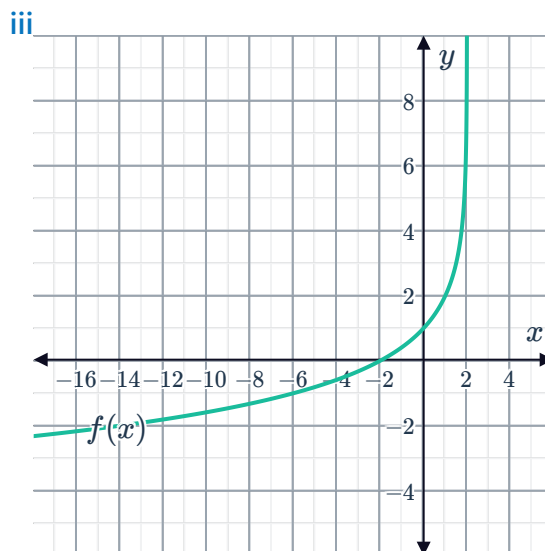
ii $x = -1$



h

i $x = -2$

ii $x = 2$





Domain and range of logarithmic functions

22

- a $f(x)$ has been translated 5 units to the right to give $g(x)$.
b The domain changes, but the range remains unchanged.

23

- a
i $y = \frac{1}{5}e^x$
ii $(-\infty, \infty)$
iii $(0, \infty)$
- b
i $y = \frac{1}{4}(e^x + 3)$
ii $(-\infty, \infty)$
iii $\left(\frac{3}{4}, \infty\right)$
- c
i $y = \frac{1}{2}\left(e^{\frac{x}{4}} + 5\right)$
ii $(-\infty, \infty)$
iii $\left(\frac{5}{2}, \infty\right)$
- d
i $y = \frac{1}{5}e^{\frac{x+3}{6}}$
ii $(-\infty, \infty)$
iii $(0, \infty)$
- e
i $y = e^{\frac{x+3}{6}} + 4$
ii $(-\infty, \infty)$
iii $(4, \infty)$
- f
i $y = \log_e x - 3$
ii $(0, \infty)$
iii $(-\infty, \infty)$
- g
i $y = \log_e (x + 3)$
ii $(-3, \infty)$
iii $(-\infty, \infty)$
- h
i $y = -\log_e (5 - x)$
ii $(-\infty, 5)$
iii $(-\infty, \infty)$

i

i $y = \log_e \left(\frac{x}{2} \right) - 3$



ii $(0, \infty)$

iii $(-\infty, \infty)$

j

i $y = \frac{1}{5} \log_e \left(\frac{x-4}{3} \right)$

ii $(4, \infty)$

iii $(-\infty, \infty)$

24

a $x > 0$

b $x < 0$

c $x > 0$

d $x < 0$

e $x > -3$

f $x > -5$

g $x < 3, x > 7$

h $-4 < x < 0, x > 4$

i $x > -5$

j $x < 3$

k $-\infty < x < \infty$

l $-5 < x < 5$